

TECHNOGLOBE – BHARATPUR CENTRE

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CURRICULUM & SYLLABUS: COURSE-BASED INTERNSHIP IN DATA ANALYTICS & BUSINESS INTELLIGENCE

Course Overview: Industry-aligned comprehensive internship covering Excel, SQL, Python, Power BI, Statistics and Capstone Analytics.

Duration: 6 Weeks | **Total Structured Hours:** 120 Hours | **Standard Delivery:** Offline

Module & Title	Core Topics & Technologies	Practical Activity & Outcomes	Hrs
Module 1 Introduction to Data Analytics	- Data Analytics Fundamentals - Types of Data (Structured vs Unstructured) - Data Analytics Lifecycle - Business Problems & Analytical Thinking - Metrics & KPIs	Practicals: - Case study evaluation on real-world retail problems - Mapping business questions to data metrics Outcomes: - Understand analytics workflow - Formulate clear business questions	12 hrs
Module 2 Microsoft Excel	- Data Cleaning & Formatting - Formulas & Mathematical Functions - Lookup Functions (XLOOKUP, VLOOKUP, INDEX-MATCH) - Pivot Tables & Pivot Charts - Interactive Dashboards - Data Validation & What-If Analysis	Practicals: - Building a dynamic sales tracker in Excel - Automated executive summary dashboard Outcomes: - Proficiency in advanced Excel functions - Create interactive business dashboards	14 hrs
Module 3 SQL	- Database Architecture & RDBMS - SELECT, WHERE, ORDER BY, GROUP BY - JOINS (INNER, LEFT, RIGHT, FULL) - Aggregations & HAVING - Subqueries & Common Table Expressions (CTEs) - Practical SQL Exercises	Practicals: - Writing SQL queries on multi-table e-commerce database - Customer RFM segmentation in SQL Outcomes: - Design and query relational schemas - Extract business insights using SQL	16 hrs
Module 4 Python for Data Analytics	- Python Fundamentals & Syntax - Variables, Data Types, Collections - Functions & Functional Logic - NumPy Arrays & Mathematical Operations - Pandas DataFrames & Series - Exploratory Data Analysis	Practicals: - Exploratory data analysis of 50,000+ transaction rows - Automated data transformation pipeline Outcomes: - Clean and reshape raw data with Pandas - Execute thorough exploratory data analysis	18 hrs
Module 5 Data Cleaning & Transformation	- Missing Value Imputation Strategies - Outlier Detection & Capping - Data Type Conversions & Date Parsing - Feature Encoding & Scaling - Data Quality Auditing	Practicals: - Cleaning real messy multi-source datasets - Building reusable data sanitization functions Outcomes: - Clean and transform messy data reliably - Ensure statistical integrity of sanitized datasets	12 hrs
Module 6 Data Visualization	- Visualization Principles & Visual Perception - Chart Selection Matrix - Color Theory & Aesthetic Layouts - Dashboard Design Thinking - Business Storytelling with Data	Practicals: - Creating publication-ready charts - Visual audit of cluttered charts and redesigning Outcomes: - Apply professional data visualization standards - Communicate insights persuasively to leadership	10 hrs

Module 7 Power BI	• Data Import & Power Query ETL • Data Transformation • Data Modeling & Relationships • DAX Fundamentals & Calculated Columns • Key Performance Indicators (KPIs) • Interactive Dashboards & Publishing	Practicals: • End-to-end interactive Power BI dashboard with drill-through • Custom KPI card suite with DAX Outcomes: • Build end-to-end Power BI solutions • Write core DAX business metrics	18 hrs
Module 8 Statistics & Business Intelligence	• Descriptive Statistics (Mean, Median, Mode) • Variance & Standard Deviation • Correlation vs Causation • Normal Distribution & Outliers • Enterprise Business KPIs	Practicals: • Statistical hypothesis testing on promotional data • Outlier detection and resolution Outcomes: • Apply statistical thinking to business metrics • Interpret variance and correlation safely	10 hrs
Module 9 Capstone Project	• Problem Formulation & Dataset Acquisition • Data Cleaning & Preparation • SQL Querying & Python Transformation • Power BI Interactive Dashboard • Executive Findings & Actionable Recommendations	Practicals: • Complete end-to-end Capstone Project Report and presentation Outcomes: • Deliver complete industry-standard analytical portfolio project	10 hrs